Lecture 9. Additional Topics in Probability

Poisson process, Markov chains, surprise and entropy, coding theory

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9.1 The Poisson process

Motivation. Random events with independent increments and no simultaneity form a Poisson process: Poisson counts, exponential gaps.

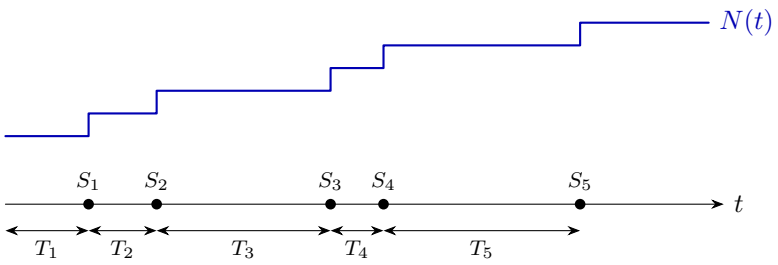


Figure. *Arrival times S_k, i.i.d.* Exp(λ) *gaps $T_k = S_k - S_{k-1}$, and the counting process $N(t) \sim$* Poisson(λt).

Definition (Poisson process)

A family $\{N(t), t \geq 0\}$ counting events in $[0, t]$ is a **Poisson process with rate** $\lambda > 0$ if (i) $N(0) = 0$;
(ii) the numbers of events in disjoint intervals are independent (independent increments);
(iii) the law of the number of events in $(s, s + t]$ depends only on t (stationary increments);
(iv) $P(N(h) = 1) = \lambda h + o(h)$ and $P(N(h) \geq 2) = o(h)$.

Proposition 9.1

For a Poisson process, $N(t) \sim$ Poisson(λt), and more generally the number of events in any interval of length t is Poisson(λt).

Proof. Split $[0, t]$ into n pieces. By (iv) and Boole, some piece has ≥ 2 events with probability $\leq n\, o(t/n) \to 0$; otherwise $N(t)$ counts pieces with an event, Bin$(n, \lambda t/n + o(t/n))$ by (ii)–(iv), with Poisson(λt) limit (Proposition 4.4). ■

Let T_1 be the first event time and T_n the n-th interarrival time.

Proposition 9.2

The interarrival times $T_1, T_2, \ldots$ are i.i.d. Exp(λ). Hence the time $S_n = T_1 + \cdots + T_n$ of the n-th event is Gamma(n, λ).

Proof. $P(T_1 > t) = P(N(t) = 0) = e^{-\lambda t}$. Given $T_1 = s$, $T_2 > t$ means no event in $(s, s + t]$, probability $e^{-\lambda t}$ for every s; so $T_2 \sim$ Exp(λ) independent of T_1; induct. The gamma law: Proposition 6.2(a), or $P(S_n \leq t) = P(N(t) \geq n)$. ■

Conversely, i.i.d. exponential gaps give a Poisson process (memorylessness yields independent increments).

Example 9.1

Earthquakes at rate 2 per week: at least 3 in 2 weeks has probability $1 - 13e^{-4} \approx 0.762$; next quake more than 2 weeks away, $e^{-4} \approx 0.018$.

Example 9.2 (Location of an event)

Given $N(t) = 1$, the time T_1 of the event is uniform on $[0, t]$: for $s \le t$,

$$P(T_1 \le s \mid N(t) = 1) = \frac{P(N(s) = 1)P(N(t) - N(s) = 0)}{P(N(t) = 1)} = \frac{\lambda s e^{-\lambda s} e^{-\lambda(t-s)}}{\lambda t e^{-\lambda t}} = \frac{s}{t}.$$

Given $N(t) = n$, the event times are distributed as n ordered i.i.d. $U(0, t)$ variables.

With Example 6.4: independently marking events as type 1 with probability p gives *independent* Poisson processes of rates λp and $\lambda(1 - p)$.

9.2 Markov chains

Motivation. Memoryless evolution: n-step probabilities are matrix powers, often converging to a stationary law.

Definition (Markov chain)

A sequence $X_0, X_1, \ldots$ with values in $\{0, 1, \ldots, M\}$ is a **Markov chain** if $P(X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \ldots, X_0 = i_0) = P_{ij}$ for all states and n, with **transition probabilities** $P_{ij} \ge 0$, $\sum_j P_{ij} = 1$.

With transition matrix $\mathbf{P} = (P_{ij})$,

$$P(X_0 = i_0, \ldots, X_n = i_n) = P(X_0 = i_0) P_{i_0 i_1} \cdots P_{i_{n-1} i_n}.$$

Example 9.3

(a) Weather: rain tomorrow w.p. α after rain, β otherwise; with 0 = rain, $\mathbf{P} = \begin{pmatrix} \alpha & 1-\alpha \\ \beta & 1-\beta \end{pmatrix}$.
(b) Gambler's ruin: $P_{i,i+1} = p = 1 - P_{i,i-1}$ $(0 < i < N)$, $P_{00} = P_{NN} = 1$ (absorbing).
(c) Ehrenfest: M molecules in two urns, a random one switches each step; with X_n in urn 1, $P_{i,i-1} = i/M$, $P_{i,i+1} = (M - i)/M$.
(d) Weather depending on two days: (yesterday, today) is a four-state chain; enlarging the state makes longer memory Markov.

Proposition 9.3 (Chapman–Kolmogorov)

The n-step transition probabilities $P_{ij}^{(n)} = P(X_{m+n} = j \mid X_m = i)$ satisfy $P_{ij}^{(n+m)} = \sum_k P_{ik}^{(n)} P_{kj}^{(m)}$; hence $(P_{ij}^{(n)}) = \mathbf{P}^n$.

Proof. Condition on the state after n steps and use the Markov property. ∎

Example 9.4

Weather with $\alpha = 0.7$, $\beta = 0.4$: $(\mathbf{P}^4)_{00} = 0.5749$, $(\mathbf{P}^4)_{10} = 0.5668$; the start is almost forgotten.

Theorem 9.4 (Limiting probabilities)

If for some n all entries of $\mathbf{P}^n$ are positive (the chain is *ergodic*), then $\pi_j = \lim_{n\to\infty} P_{ij}^{(n)}$ exists, does not depend on i, and is the unique solution of

$$\pi_j = \sum_i \pi_i P_{ij} \quad (j = 0, \dots, M), \qquad \sum_j \pi_j = 1.$$

Let $n \to \infty$ in $P_{ij}^{(n+1)} = \sum_k P_{ik}^{(n)} P_{kj}$ (existence quoted). π is **stationary** ($X_0 \sim \pi$ gives $X_n \sim \pi$), π_j is the long-run fraction of time in j, and $1/\pi_j$ the mean return time.

Example 9.5

(a) Weather: $\pi_0 = \alpha\pi_0 + \beta\pi_1$, $\pi_0 + \pi_1 = 1$ give $\pi_0 = \beta/(1+\beta-\alpha) = 4/7 \approx 0.571$ (cf. Example 9.4).
(b) Ehrenfest: $\pi_j = \binom{M}{j}2^{-M}$ satisfies $\pi_{j-1}\frac{M-j+1}{M} + \pi_{j+1}\frac{j+1}{M} = 2^{-M}\{\binom{M-1}{j-1} + \binom{M-1}{j}\} = \pi_j$, so each molecule is in either urn w.p. $\frac{1}{2}$, independently. (Period 2: $P_{ij}^{(n)}$ does not converge, but π_j is still the long-run fraction.)

9.3 Surprise, uncertainty and entropy

The surprise $S(p)$ of an event of probability p, if decreasing and continuous with $S(1) = 0$ and $S(pq) = S(p) + S(q)$, must be $-C\log_2 p$ (the equation of Proposition 5.4); take $C = 1$ (**bits**).

The **entropy** of X with probabilities $p_1, \dots, p_n$ is the expected surprise

$$H(X) = -\sum_i p_i \log_2 p_i,$$

It is 0 iff X is constant and $\le \log_2 n$, with equality iff uniform (Jensen: $\sum p_i \log(1/p_i) \le \log \sum p_i \cdot (1/p_i)$).

The joint entropy is $H(X,Y) = -\sum_{i,j} p(x_i, y_j)\log_2 p(x_i, y_j)$, and the conditional entropy $H_Y(X) = \sum_j H(X \mid Y = y_j)P(Y = y_j)$ is the uncertainty left once Y is known.

Proposition 9.5

(a) $H(X,Y) = H(Y) + H_Y(X)$.
(b) $H_Y(X) \le H(X)$, with equality iff X and Y are independent.

Proof. (a) $\log p(x_i, y_j) = \log p(y_j) + \log p(x_i \mid y_j)$; take $-E$.

(b) With $\log x \le x - 1$,

$$H_Y(X) - H(X) = \sum_{i,j} p(x_i, y_j)\log_2 \frac{p(x_i)}{p(x_i \mid y_j)} \le \frac{1}{\ln 2}\sum_{i,j} p(y_j)\{p(x_i) - p(x_i \mid y_j)\} = 0,$$

with equality iff $p(x_i \mid y_j) = p(x_i)$ whenever $p(x_i, y_j) > 0$. ∎

So observing Y never increases uncertainty on average; $H(X) - H_Y(X)$ is the information Y carries about X.

9.4 Coding theory and entropy

Transmit X (values x_i, probabilities p_i) as binary strings decodable in sequence. In a **prefix code** no codeword begins another, so concatenations decode by reading until a codeword ends.

Lemma 9.6 (Kraft's inequality)

A prefix code with codeword lengths $n_1, \dots, n_N$ exists iff $\sum_i 2^{-n_i} \le 1$.

Proof. With $L = \max n_i$, a codeword of length n_i prefixes 2^{L-n_i} strings of length L, disjointly for a prefix code: $\sum_i 2^{L-n_i} \le 2^L$. Conversely, assign codewords by increasing length, avoiding extensions of earlier ones; the count shows one is always available. ■

Theorem 9.7 (Noiseless coding theorem)

Every prefix code for X has expected length $\sum_i p_i n_i \ge H(X)$, and there is a prefix code with expected length $< H(X) + 1$.

Proof. With $q_i = 2^{-n_i} / \sum_k 2^{-n_k}$, as in Proposition 9.5(b), $-\sum p_i \log_2 p_i \le -\sum p_i \log_2 q_i = \sum p_i n_i + \log_2 \sum_k 2^{-n_k} \le \sum p_i n_i$ (Kraft). Upper bound: $n_i = \lceil \log_2(1/p_i) \rceil$ has $\sum 2^{-n_i} \le 1$ and $n_i < \log_2(1/p_i) + 1$. ■

Coding blocks of k copies gets within $1/k$ of $H(X)$ per symbol: entropy is the minimal bits per symbol.

Example 9.6

Probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}$: $H(X) = 1.75$ bits, attained by $0, 10, 110, 111$; a fixed-length code needs 2.

Noisy channels. If each bit flips independently w.p. p, Shannon's theorem: rates below the **capacity** $C = 1 + p \log_2 p + (1-p) \log_2(1-p) = 1 - H(\text{Bernoulli}(p))$ admit codes with error probability $\to 0$; rates above C do not.

9.5 Selected exercises

Motivation. Ross's problems and exercises (book numbering): a Poisson computation, a stationary law by symmetry, and entropy chain-rule consequences.

Exercise 9.1 (Two arrivals in the first hour)

Customers arrive at a bank as a Poisson process with rate λ. Given that two customers arrived during the first hour, find the probability that (a) both arrived during the first 20 minutes; (b) at least one did.

Solution.

Step 1. In hours, $N(1/3)$ and $N(1) - N(1/3)$ are independent Poisson$(\lambda/3)$, Poisson$(2\lambda/3)$, and $N(1) \sim$ Poisson(λ) (Proposition 9.1).

Step 2. For $k = 0, 1, 2$,

$$P(N(1/3) = k \mid N(1) = 2) = \frac{e^{-\lambda/3}\frac{(\lambda/3)^k}{k!}\, e^{-2\lambda/3}\frac{(2\lambda/3)^{2-k}}{(2-k)!}}{e^{-\lambda}\lambda^2/2} = \binom{2}{k}\left(\frac{1}{3}\right)^k\left(\frac{2}{3}\right)^{2-k},$$

$\mathrm{Bin}(2, \frac{1}{3})$, free of λ (the arrivals act as independent uniform times).

(a) $P(N(1/3) = 2 \mid N(1) = 2) = 1/9$.

(b) $P(N(1/3) \geq 1 \mid N(1) = 2) = 1 - (2/3)^2 = 5/9$. ■

Exercise 9.7 (Doubly stochastic chains)

A transition matrix on $\{0, 1, \ldots, M\}$ is doubly stochastic if $\sum_{i=0}^{M} P_{ij} = 1$ for every j. Show that if such a chain is ergodic, then $\pi_j = 1/(M+1)$ for all j.

Solution.

Step 1. $\pi_j = 1/(M+1)$ has $\sum_j \pi_j = 1$ and

$$\sum_{i=0}^{M} \pi_i P_{ij} = \frac{1}{M+1}\sum_{i=0}^{M} P_{ij} = \frac{1}{M+1} = \pi_j.$$

Step 2. By uniqueness in Theorem 9.4, these are the limiting probabilities. ■

Exercise 9.17 (Functions do not increase entropy)

Show that $H(f(X)) \leq H(X)$ for every discrete X and every function f.

Solution.

Step 1. With $Y = f(X)$, $(X, Y) = (x, f(x))$ w.p. $P(X = x)$, so $H(X, Y) = H(X)$.

Step 2. $H(X, Y) = H(Y) + H_Y(X)$ (Proposition 9.5(a)) and $H_Y(X) \geq 0$.

Step 3. So $H(Y) \leq H(X)$, with equality iff $H_Y(X) = 0$, i.e. f is one-to-one on the values of X. ■

Exercise 9.18 (Capacity of the binary symmetric channel)

A bit X is sent and Y received, with $P(Y = 1 \mid X = 1) = P(Y = 0 \mid X = 0) = p$. The rate of transmission is $H(X) - H_Y(X)$ and the capacity is its maximum over $\alpha = P(X = 1)$. Show that the maximum is attained at $\alpha = \frac{1}{2}$ and equals $1 + p\log_2 p + (1-p)\log_2(1-p)$.

Solution.

Step 1. $H(X) + H_X(Y) = H(X, Y) = H(Y) + H_Y(X)$, so the rate is $H(Y) - H_X(Y)$.

Step 2. With $h(p) = -p\log_2 p - (1-p)\log_2(1-p)$, $H(Y \mid X = x) = h(p)$ for both x, so $H_X(Y) = h(p)$ for every α.

Step 3. $H(Y) \leq 1$, with equality iff $P(Y = 1) = \alpha p + (1-\alpha)(1-p) = \frac{1}{2}$, e.g. $\alpha = \frac{1}{2}$.

Step 4. So the capacity is $1 - h(p)$, the C of Section 9.4. ■